



# Air Quality Situation Report

Edmonton Symphony Orchestra Outdoor Concert · Four Seasons Park

Sun, Aug 23, 2026 · 3:14 PM MDT

## Air Quality Health Index (AQHI) — Edmonton

# 4.0

Current AQHI at Edmonton Lendrum (18.14 km from venue)  
Forecast Sun 5:00 PM at Edmonton Lendrum (18.14 km from venue): **4.3**

| Health Risk   | AQHI | At-Risk Population                                                                                | General Population                                                                                                     |
|---------------|------|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Moderate Risk | 4.3  | Consider reducing or rescheduling strenuous activities outdoors if you are experiencing symptoms. | No need to modify your usual outdoor activities unless you experience symptoms such as coughing and throat irritation. |

No active Environment Canada weather alerts for the venue.

## Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 23.3°C and feels near 23.4°C. Winds are 15 km/h from W, gusting near 30 km/h.
- The nearest current AQHI is 4 at Edmonton Lendrum, 18.1 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 5 (forecast tonight: 4, tomorrow: 2).
- A gridded AQHI estimate blending official and community sensors (confidence: High, nearest contributing point 6.1 km away) puts the area near 4.
- The nearest air monitoring station (Leduc Sensor, 12.7 km) reports fine particulate matter (PM2.5) at 43.4 µg/m³.
- A nearby community sensor ('ACA\_RMA', 7.4 km) reads 41.1 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 51.2 km to the NW, i.e. arriving from the NW.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-08-22) is 20.8 km N of the venue. The modeled back-trajectory does not pass near this fire (nearest approach 21.0 km), so this specific fire is unlikely to be the source of any smoke reaching the venue right now.
- The AQHI forecast for Sun 5:00 PM is 4, based on Edmonton Lendrum, 18.1 km from the venue.
- Thunderstorm conditions appear in the hourly forecast beginning around Sun 8:00 PM.
- Overall operational environmental risk is HIGH. Primary concerns are precipitation, thunderstorm.

## Recommendations

- Confirm lightning monitoring, shelter, pause and evacuation procedures.
- Prepare drainage, electrical protection and wet-weather controls.
- Environment Canada AQHI guidance — general population: No need to modify your usual outdoor activities unless you experience symptoms such as coughing and throat irritation.
- Environment Canada AQHI guidance — at-risk populations: Consider reducing or rescheduling strenuous activities outdoors if you are experiencing symptoms.

## Hazard Snapshot — Overall Risk: High Risk

|                                                                     |                                                          |                                                |                                                                     |                                                                       |
|---------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------|
| AIR QUALITY<br><b>Moderate Risk</b><br>4.3 AQHI                     | HEAT<br><b>Moderate Risk</b><br>28.1 °C apparent/humidex | WIND<br><b>Low Risk</b><br>37.4 km/h peak gust | PRECIPITATION<br><b>High Risk</b><br>3.6 mm/h maximum               | THUNDERSTORM<br><b>High Risk</b><br>Sun 8:00 PM first forecast signal |
| AQHI RATE OF CHANGE<br><b>Moderate Risk</b><br>1.0 AQHI change/hour |                                                          |                                                | WIND SHEAR<br><b>Low Risk</b><br>5.7 ° direction diff (10m vs 120m) |                                                                       |

## Local Monitoring

|                                                              |                                                   |                                                                 |                                                            |
|--------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------|
| Nearest station<br><b>4.0</b><br>Edmonton Lendrum · 18.14 km | Blended estimate<br><b>4.0</b><br>confidence High | PM2.5 (station)<br><b>43.4 µg/m³</b><br>Leduc Sensor · 12.73 km | PM2.5 (community)<br><b>41.1 µg/m³</b><br>ACA_RMA · 7.4 km |
|--------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------|

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Prepared by Alberta Capital Airshed for Edmonton Symphony Orchestra outdoor concerts. Beta decision-support product  
— for the interactive map, live cameras and hourly outlook, scan the QR code or visit [https://dkevinm.github.io/](https://dkevinm.github.io/eso_sitrep/)

